

Day 1

Electromagnetic Theory

Applications:-

→ Glucose meter, transformers, Motors, ICs, Wireless, Satellites, mobile phone, Bullet train,

Gauss's law for magnetism:-

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\oint_S \vec{D} \cdot d\vec{s} = Q_{enc}$$

Faraday's law

$$\vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt}$$

$$\oint_C \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_S \vec{B} \cdot \vec{n} \, dA$$

Ampere's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{j} \cdot d\vec{A}$$

Scalar

- Only magnitude
- Uni dimensional

→ Scalars can be divided by another scalar

→ Time, mass, length

Example

Vector

- Magnitude & direction
- More than one dimension

→ Vectors cannot be divided by another vector

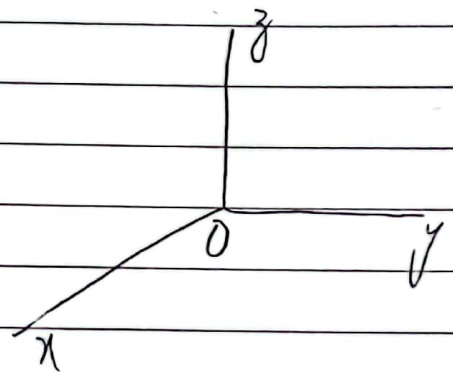
→ Velocity, Weight, Force

Basis of vector:-

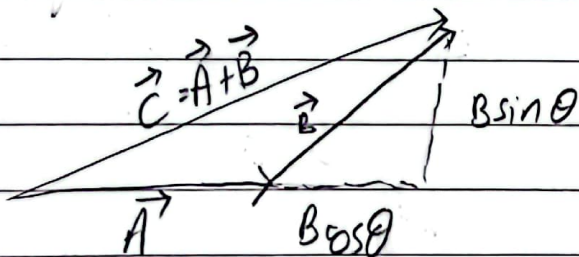
$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \hat{i}_A$$



∴ Vector Addition (head to tail rule)
(11grom)



$$C^2 = (A + B \cos \theta)^2 + (B \sin^2 \theta)$$

$$C^2 = A^2 + B^2 \cos^2 \theta + 2AB \cos \theta + B^2 \sin^2 \theta$$

$$C^2 = A^2 + 2AB \cos \theta + B^2 (\cos^2 \theta + \sin^2 \theta)$$

$$C^2 = A^2 + B^2 + 2AB \cos \theta$$

Dot product:-

$$\vec{C} = \vec{A} \cdot \vec{B}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$= A_x B_x + A_y B_y + A_z B_z$$

Cross Product

(To find direction use thumb rule) $\therefore$ right hand rule

$$\vec{Y} = \vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{Y} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

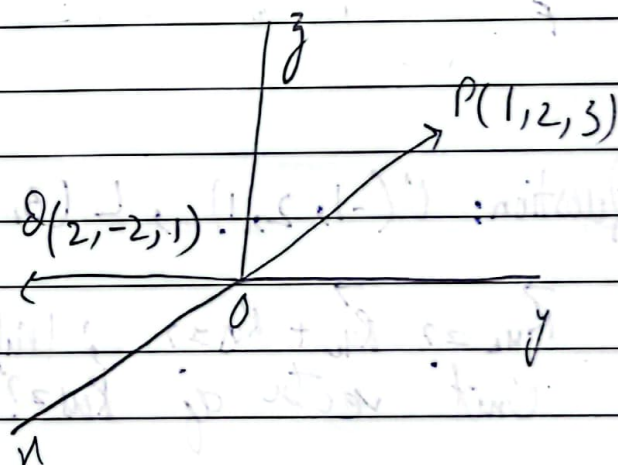
Example:-

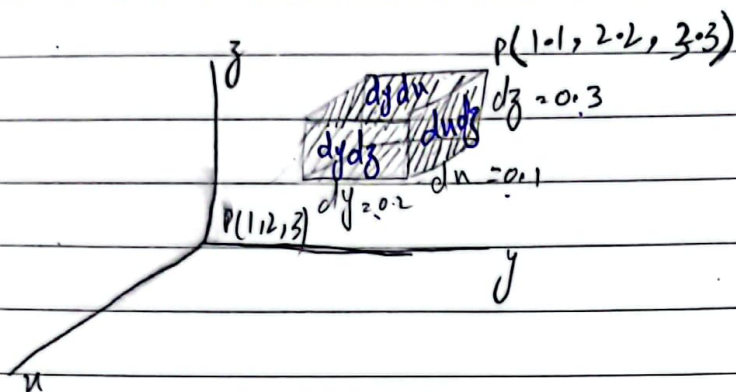
$$\vec{r}_{PO} = (1\hat{i}, -4\hat{j}, -2\hat{k})$$

$$|\vec{r}_{PO}| = \sqrt{(1)^2 + (-4)^2 + (-2)^2}$$

$$= \sqrt{1+16+4}$$

$$|\vec{r}_{PO}| = \sqrt{21}$$





$$dV = dx dy dz$$

$$|dL| = \sqrt{dx^2 + dy^2 + dz^2}$$

distance between the origin & the moved point

Question: Specify the unit vector from the origin towards the point $F(2, -2, 1)$

$$\vec{F} = (2, -2, 1)$$

$$|\vec{F}| = \sqrt{2^2 + (-2)^2 + (1)^2}$$

$$= \sqrt{4 + 4 + 1}$$

$$= \sqrt{9}$$

$$|\vec{F}| = 3$$

$$\hat{F} = \frac{(2\hat{i}, -2\hat{j}, \hat{k})}{3}$$

$$\hat{F} = \left(\frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{1}{3}\hat{k} \right)$$

Question: $U(-1, 2, 1)$, $L(3, -3, 0)$, $S(-2, -3, -4)$

$$\vec{R}_{UL} = ? \quad \vec{R}_{UL} + \vec{R}_{US} = ? \quad ; \quad |\vec{R}_U| = ? \quad (|\vec{R}_S|/2)$$

Unit vector of $\vec{R}_{UL} = ?$

$$R_{ul} = R_l - R_u$$

$$\vec{R}_{ul} = 4\hat{i} + (-5)\hat{j} - 1\hat{k}$$

$$R_{us} = R_s - R_l$$

$$\vec{R}_{us} = -1\hat{i} - 5\hat{j} - 5\hat{k}$$

$$\boxed{\vec{R}_{ul} + \vec{R}_{us} = 3\hat{i} - 10\hat{j} - 6\hat{k}}$$

$$|R_u| = \sqrt{(-1)^2 + (-2)^2 + (1)^2}$$

$$= \sqrt{1+4+1}$$

$$= \sqrt{6}$$

$$|R_s| = \sqrt{(-2)^2 + (-3)^2 + (-4)^2}$$

$$= \sqrt{4+9+16}$$

$$= \sqrt{29}$$

$$|R_{us}| = \sqrt{(-1)^2 + (-5)^2 + (-5)^2}$$

$$= \sqrt{1+25+25}$$

$$\boxed{|R_{us}| = \sqrt{51}}$$

$$\hat{i}_{us} = \frac{-1}{\sqrt{51}}\hat{i} - \frac{5}{\sqrt{51}}\hat{j} - \frac{5}{\sqrt{51}}\hat{k}$$

Cylindrical co-ordinate system:-

Real world Application:-

- Electromagnetics
- Fluid Dynamics
- Heat conduction in cycles
- Electromagnetic field in wires
- Flow in pipes

Rectangular coordinate system

cylindrical coordinate system

→ Define by ρ, ϕ and z

→ Define by x, y and z axis

→ ρ (radial distance from the z -axis)

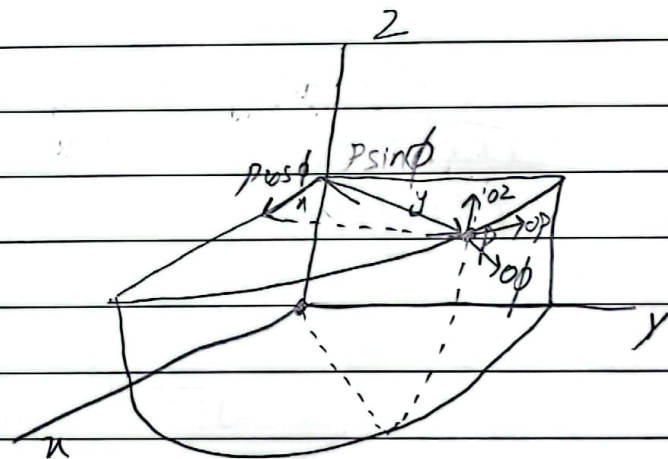
→ ϕ (Angle it makes in the xy plane)

→ z (distance along z axis)

cylindrical co-ordinates

→ surface of constant ρ are centered along z -axis

→ surface of constant ϕ are planes passing through z -axis



→ Surface of constant z are horizontal planes.

∴ ρ is always the same throughout the same cylinder.

∴ unit vector of ρ is tangent

Conversion of from Rectangular co-ordinate system to cylindrical co-ordinate system.

$$\rightarrow \rho = \sqrt{x^2 + y^2}$$

→ x, y, z given

$$\rightarrow \tan \phi = \frac{y}{x} \Rightarrow \phi = \tan^{-1} \frac{y}{x}$$

$$\rightarrow \phi = \tan^{-1} \frac{y}{x}$$

$$\rightarrow z = z$$

Conversion from CCS to RCS:-

→ ρ, ϕ, z given.

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

Unit vector

→ a_ρ, a_ϕ, a_z (CCS)

→ a_x, a_y, a_z (RCS)

Transforming unit vector:-

Dot Product

	a_p	a_ϕ	a_z
a_n	$\cos\phi$	$-\sin\phi$	0
a_y	$\sin\phi$	$\cos\phi$	0
a_z	0	0	1

$$A = A_p a_p + A_\phi a_\phi + A_z a_z$$

$$A = A_n a_n + A_y a_y + A_z a_z$$

$$A_{rect} = A_n a_n + A_y a_y + A_z a_z$$

$$A_{cyl} = A_p a_p + A_\phi a_\phi + A_z a_z$$

→ To find any desired component of a vector, we take the dot product of the vector and a unit vector in desired component

$$A_p = A_{rect} \cdot a_p$$

$$A_\phi = A_{rect} \cdot a_\phi$$

$$A_z = A_z$$

$$A_p = A_{rect} \cdot a_p = (A_x a_x + A_y a_y + A_z a_z) \cdot a_p$$

$$A_p = A_x(a_x \cdot a_p) + A_y(a_y \cdot a_p) + A_z(a_z \cdot a_p)$$

$$A_p = A_x \cos \phi + A_y \sin \phi + A_z (0)$$

$$A_p = A_x \cos \phi + A_y \sin \phi$$

$$A_\phi = A_{rect} \cdot a_\phi = (A_x a_x + A_y a_y + A_z a_z) \cdot a_\phi$$

$$A_\phi = A_x(a_x \cdot a_\phi) + A_y(a_y \cdot a_\phi) + A_z(a_z \cdot a_\phi)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi + A_z (0)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$A_z = A_z$$

$$B = y_{ax} - n_{ay} + z_{az}$$

Transform it into CCS

$$B_{cyl} = B_p a_p + B_\phi a_\phi + B_z a_z$$

$$B_p = B \cdot a_p = (y_{ax} - n_{ay} + z_{az}) \cdot a_p$$

$$B_\phi = B \cdot a_\phi = (y_{ax} - n_{ay} + z_{az}) \cdot a_\phi$$

$$B\rho = y \cos \phi - x \sin \phi$$

$$B\phi = -y \sin \phi - x \cos \phi$$

Assesment :-

Example # 1.3

Transform $B = y a_x - x a_y + z a_z$ into cylindrical co-ordinates.

Solution:-

$$B_{cyl} = B_{\rho} \hat{\rho} + B_{\phi} \hat{\phi} + B_z \hat{z}$$

$$B_{\rho} = (y a_x - x a_y + z a_z) \cdot \hat{\rho}$$

$$= y(a_x \cdot \hat{\rho}) - x(a_y \cdot \hat{\rho}) + z(a_z \cdot \hat{\rho})$$

$$= y \cos \phi - x \sin \phi + z(1)$$

$$= y \cos \phi - x \sin \phi$$

putting value of "y" & "x"

$$= \rho \sin \phi \cos \phi - \rho \cos \phi \sin \phi$$

$$= 0$$



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Date: / /

$$B\phi = (y a_u - u a_y + z a_z) \cdot a\phi$$

$$= y(a_u \cdot a\phi) - u(a_y \cdot a\phi) + z(a_z \cdot a\phi)$$

$$= y(1 - \sin\phi) - u \cos\phi + z(0)$$

$$= -y \sin\phi - u \cos\phi$$

put value of "y" & "u"

$$= -p \sin^2\phi - p \cos^2\phi$$

$$= -p(\sin^2\phi + \cos^2\phi)$$

$$= -p(1)$$

$$\boxed{B\phi = -p}$$

Spherical co-ordinate system:-

RCS $\rightarrow x, y, z$

CCS $\rightarrow \rho, \phi, \theta$

SCS $\rightarrow r, \phi, \theta$

- $\rightarrow r \rightarrow$ the distance from origin to the point
- $\rightarrow$ Polar Angle (θ) $\rightarrow$ The angle between the z-axis and the line connecting to the region.
- $\rightarrow$ Azimuthal Angle (ϕ) $\rightarrow$ The angle between the x-axis and projection of the line connecting the origin to the point onto the xy-plane.

RCS $\rightarrow$ 3 planes

CCS $\rightarrow$ 2 planes, 1 curve

SCS $\rightarrow$ A sphere, cone, plane.

Unit vector:-

a_r, a_θ, a_ϕ

- $a_r \rightarrow$ Radial unit vector, directed outward from the origin, normal to the sphere
- $a_\theta \rightarrow$ tangent to the surface and pointing in the direction of increasing θ , normal to the conical plane.
- $a_\phi \rightarrow$ tangent to the sphere & cone, pointing in the direction of increasing ϕ .

Transformation from RCS to SCS
 $(x, y, z) \rightarrow (r, \theta, \phi)$

$$r = \sqrt{x^2 + y^2 + z^2} \quad (r \geq 0)$$

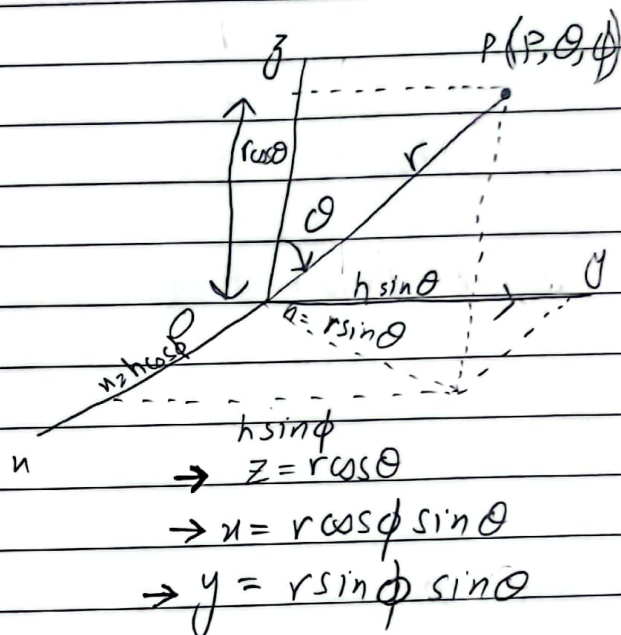
$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \quad (0 \leq \phi \leq 360)$$

$$z = r \cos \theta$$

$$\cos \theta = \frac{z}{r}$$

$$\cos \theta = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \quad (0 \leq \theta \leq 180^\circ)$$



Transformation of vectors:-

The transformation of vectors requires determining the dot product of spherical unit vector with rectangular unit vectors.

Purpose of dot product:-

To find the component of a vector in the direction of a new unit vector (from SCS), we take the dot product of original vector's (from RCS) with the new vectors.

Mathematical Representation:-

In spherical co-ordinates system

$$A_{\text{sphere}} = A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi}$$

The components of the vector A_{sphere} along the sphere co-ordinate axis

$$A_r = A_{\text{sphere}} \cdot \hat{r}$$

$$A_\theta = A_{\text{sphere}} \cdot \hat{\theta}$$

$$A_\phi = A_{\text{sphere}} \cdot \hat{\phi}$$

$$A_r = A_x (\hat{a}_x \cdot \hat{r}) + A_y (\hat{a}_y \cdot \hat{r}) + A_z (\hat{a}_z \cdot \hat{r})$$

Dot product that give the projection of the rectangular unit vectors onto the spherical unit " $\hat{r}$ ".